UNIVERSITY OF CAMBRIDGE

PBT1/NS3

NATURAL SCIENCE TRIPOS Part 1B

NST1 NB1
PBT1 NS3

Neurobiology

Thursday 1st June 2023 (13.30 – 16.30 hrs)

Answer any **4** questions.

You have **3 hours** (plus any pre-agreed individual adjustment) to answer this paper.

Diagrams or other handwritten material:

Where appropriate (e.g. for formulae, equations or diagrams) you may hand write your answers using the Inspera Scan Paper provided.

Candidates should not copy and paste images from any other documents.

Note:

- Inspera Scan question codes **MUST** be noted during the exam.
- **The Inspera Scan code for each question is found BELOW the text box for that question.**
- Please ensure that each Inspera Scan page is labelled appropriately, including your Blind Grade Number.
- Inspera Scan pages **MUST** be placed in the Inspera Scan folder at the completion of the exam, and left on your desk.
- Inspera scan pages will be scanned by administrators after the completion of the exam.

Stationery requirements:

- Rough work pad
- Inspera Scan pages
- Inspera Scan folder

Candidates are permitted to use an approved calculator.

Please turn over.

1. What have modern advances in Cryo-EM, and the predictions of 3D structures from amino acid sequences, told us about GPCR structure? What are their limitations?

2. Compare and contrast the main ascending tactile and nociceptive pathways from the skin to the cortex. How can information carried by these pathways be modified?

3. Critically evaluate the evidence for ventral and dorsal streams in visual processing.

4. Discuss the role and mode of action of retinoic acid in specifying the antero-posterior axis of the neural tube.

5. Compare and contrast the transduction, encoding and processing of chemosensory signals in olfaction and taste.

6. To what extent can the function of motor system components be considered in isolation?

7. Discuss the properties of the NMDA receptor that make it an attractive candidate for development and learning-related plasticity in the brain.

8. Discuss mechanisms that contribute to the precision of connectivity during the development of neuronal (e.g., somatotopic) maps. Give examples to illustrate your answer.

9. How did the 'lateral hypothalamic syndrome', produced by electrolytic lesions, influence our understanding of motivated behaviour?

10. What is the evidence that the hippocampus is necessary for encoding relationships between stimuli?